

Collateral Assessment

US-801 Retrieve Collateral Information

Epic: AI-Enabled Loan Origination System

User Story

As the Loan Origination System,

I want to retrieve collateral details submitted by the customer or already available in the bank's records,

So that collateral adequacy and eligibility can be assessed before the credit recommendation is generated.

Business Rules

- Collateral shall be mandatory for secured loan products.
 - Multiple collateral assets may be linked to a single loan application.
 - Collateral types shall follow bank policy.
 - Existing collateral information shall be reused whenever available.
 - Missing mandatory documents shall prevent progression.
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Acceptance Criteria

Scenario 1

Given collateral information already exists

When the assessment starts

Then

- Retrieve collateral details
 - Retrieve ownership information
 - Retrieve valuation reports
 - Proceed to collateral verification
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Scenario 2

Given collateral information is unavailable

When assessment begins

Then

- Request customer upload
 - Notify Relationship Manager
 - Mark status as Pending
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US-802 Verify Collateral Documents

User Story

As the Loan Origination System,

I want to verify ownership and supporting collateral documents,

So that only legally valid collateral is considered during credit appraisal.

Checks include

- Title Deed
 - Sale Deed
 - Encumbrance Certificate
 - Insurance
 - RC Book
 - Gold Ownership
 - Valuation Report
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US-803 AI Collateral Assessment

This is where your project becomes interesting.

User Story

As the AI Collateral Assessment Engine,

I want to analyze collateral characteristics, market value, ownership information, and policy compliance,

So that the Credit Officer receives a structured collateral risk assessment to support lending decisions.

AI performs

- Loan-to-Value calculation
- Collateral Coverage
- Asset Category

- Risk Rating
 - Depreciation
 - Liquidity
 - Concentration Risk
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US-804 AI Evaluation & Governance

As the AI Evaluation & Governance Engine,

I want to evaluate the trustworthiness of the AI-generated collateral assessment,

So that only reliable AI outputs are used during collateral evaluation.

Evaluate

- Confidence
 - Explainability
 - Missing Property Data
 - OCR Quality
 - Valuation Completeness
 - Bias
 - Hallucination
 - Policy Compliance
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US-805 Review Collateral Assessment Requiring Governance Validation

User Story

As the Loan Origination System,

I want to automatically route collateral assessments for manual review whenever the AI Evaluation & Governance Engine identifies governance concerns such as low confidence, missing valuation data, property ownership mismatch, hallucinated conclusions, or insufficient explainability,

So that a Credit Officer can validate the collateral assessment before it is used in the credit decision-making process, ensuring regulatory compliance, risk mitigation, and responsible use of AI.

Acceptance Criteria

Scenario 1: Low AI Confidence

Given the AI Collateral Assessment has been completed,

When the AI Confidence Score is below the configured threshold,

Then

- The collateral assessment shall be routed for manual review.
 - The Credit Officer shall be notified.
 - The AI recommendation shall not be used for automated workflow progression.
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Scenario 2: Missing Valuation Information

Given the collateral assessment is performed,

When mandatory valuation information is missing or incomplete,

Then

- The application shall be flagged for manual review.
 - The Credit Officer shall verify the valuation details.
 - Additional valuation documents may be requested before proceeding.
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Scenario 3: Property Ownership Mismatch

Given ownership verification has been completed,

When inconsistencies are identified between the declared owner and the supporting ownership documents,

Then

- The collateral assessment shall be routed for manual verification.
 - The ownership discrepancy shall be recorded in the audit log.
 - The application shall not proceed until the discrepancy is resolved.
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Scenario 4: AI Hallucination or Unsupported Conclusions Detected

Given the AI Evaluation & Governance Engine has completed its assessment,

When unsupported conclusions, fabricated information, or inconsistent collateral analysis are detected,

Then

- The AI-generated assessment shall be rejected.
 - The application shall be routed for manual collateral assessment.
 - The governance exception shall be recorded for audit and monitoring purposes.
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Scenario 5: Explainability Unavailable

Given the AI has generated a collateral assessment,

When the AI cannot provide sufficient reasoning or supporting evidence for its recommendation,

Then

- The application shall be routed for manual review.
 - The explainability issue shall be recorded in the governance log.
 - The Credit Officer shall perform an independent collateral assessment.
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Scenario 6: Governance Evaluation Passed

Given the AI Evaluation & Governance Engine has completed its assessment,

When

- The AI Confidence Score meets the configured threshold,
- No material bias or governance concerns are detected,
- Mandatory collateral and valuation information is complete,
- Property ownership is successfully verified,
- No hallucinations are identified, and
- Explainability requirements are satisfied,

Then

- The collateral assessment shall be marked as **Governance Approved**.
 - The assessment, governance results, and audit trail shall be stored.
 - The application shall automatically proceed to the **Credit Risk Assessment** stage.
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Business Value

This user story demonstrates that **AI recommendations do not automatically determine lending decisions**. Instead, the system enforces governance controls by requiring human oversight whenever AI outputs fail predefined trust, data quality, or policy validation checks. This approach supports **Responsible AI**, strengthens regulatory compliance, improves auditability, and ensures that collateral decisions remain transparent and accountable within the loan origination process.

US-806 Store Collateral Assessment Results

User Story

As the Loan Origination System,

I want to securely store the validated collateral assessment results, including valuation details, Loan-to-Value (LTV) ratio, collateral risk rating, AI-generated assessment, governance evaluation outcome, and audit information,

So that the collateral assessment is available for credit risk evaluation, lending decisions, regulatory compliance, and future reference throughout the loan lifecycle.

Acceptance Criteria

Scenario 1: Successful Storage

Given the collateral assessment has been completed

And the AI Evaluation & Governance process has been successfully executed

When the collateral assessment is approved

Then

- The collateral valuation shall be stored.
 - The Loan-to-Value (LTV) ratio shall be calculated and stored.
 - The collateral risk rating shall be recorded.
 - The AI-generated collateral assessment shall be stored.
 - The AI Governance Evaluation results shall be recorded.
 - The governance approval status shall be stored.
 - The assessment shall be linked to the corresponding Loan Application ID.
 - The application shall proceed to the Credit Risk Assessment stage.
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Scenario 2: Audit Trail

Given the collateral assessment has been stored

When the transaction is completed

Then

The audit log shall capture

- Assessment Date & Time
- AI Model Version
- Confidence Score
- Governance Status
- Credit Officer Review Status

- User/System ID
 - Version History
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Business Rules

- Only approved collateral assessments shall be stored.
 - All assessment records shall be linked to the Loan Application ID.
 - Assessment records shall be version controlled.
 - Audit logs shall be immutable.
 - Role-based access shall be enforced.
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US-807 Store Legal Due Diligence Results

User Story

As the Loan Origination System,

I want to securely store the legal due diligence results associated with the collateral,

So that legal verification findings are available for credit appraisal, mortgage creation, regulatory compliance, audit purposes, and future reference throughout the loan lifecycle.

Acceptance Criteria

Scenario 1: Successful Legal Assessment Storage

Given the legal due diligence has been completed

When the legal opinion is approved

Then

- The legal opinion shall be stored.
 - Ownership verification status shall be recorded.
 - Encumbrance verification status shall be stored.
 - Litigation status shall be recorded.
 - Mortgage eligibility status shall be stored.
 - Property document verification results shall be stored.
 - Legal reviewer remarks shall be stored.
 - The legal assessment shall be linked to the corresponding Loan Application ID.
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Scenario 2: Legal Exception

Given legal verification has been completed

When title defects, encumbrances, litigation, or adverse legal findings are identified

Then

- The legal exception shall be recorded.
 - The application shall be flagged for manual review.
 - The Credit Officer shall be notified.
 - Loan processing shall not proceed until the legal issue is resolved.
 - The governance exception shall be captured in the audit log.
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Scenario 3: Audit Trail

Given the legal assessment has been stored

When the transaction is completed

Then

The audit log shall capture

- Assessment Date & Time
 - Legal Reviewer
 - Legal Opinion Status
 - Ownership Verification Status
 - Encumbrance Status
 - Litigation Status
 - User/System ID
 - Version History
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Business Rules

- Legal due diligence shall be mandatory for all secured loan products.
 - Only legally verified collateral shall proceed to Credit Risk Assessment.
 - Legal opinions shall be obtained from authorized legal professionals.
 - Legal records shall be immutable after approval.
 - Any modification shall create a new version while preserving historical records.
 - Access to legal documents shall be governed through role-based access control.
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Business Value

The system securely stores legal due diligence findings, ensuring that collateral accepted by the bank has a **clear legal title, verified ownership, acceptable encumbrance status, and**

enforceable security interest. This supports informed credit decisions, mortgage creation, regulatory compliance, audit readiness, and complete traceability throughout the loan lifecycle



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